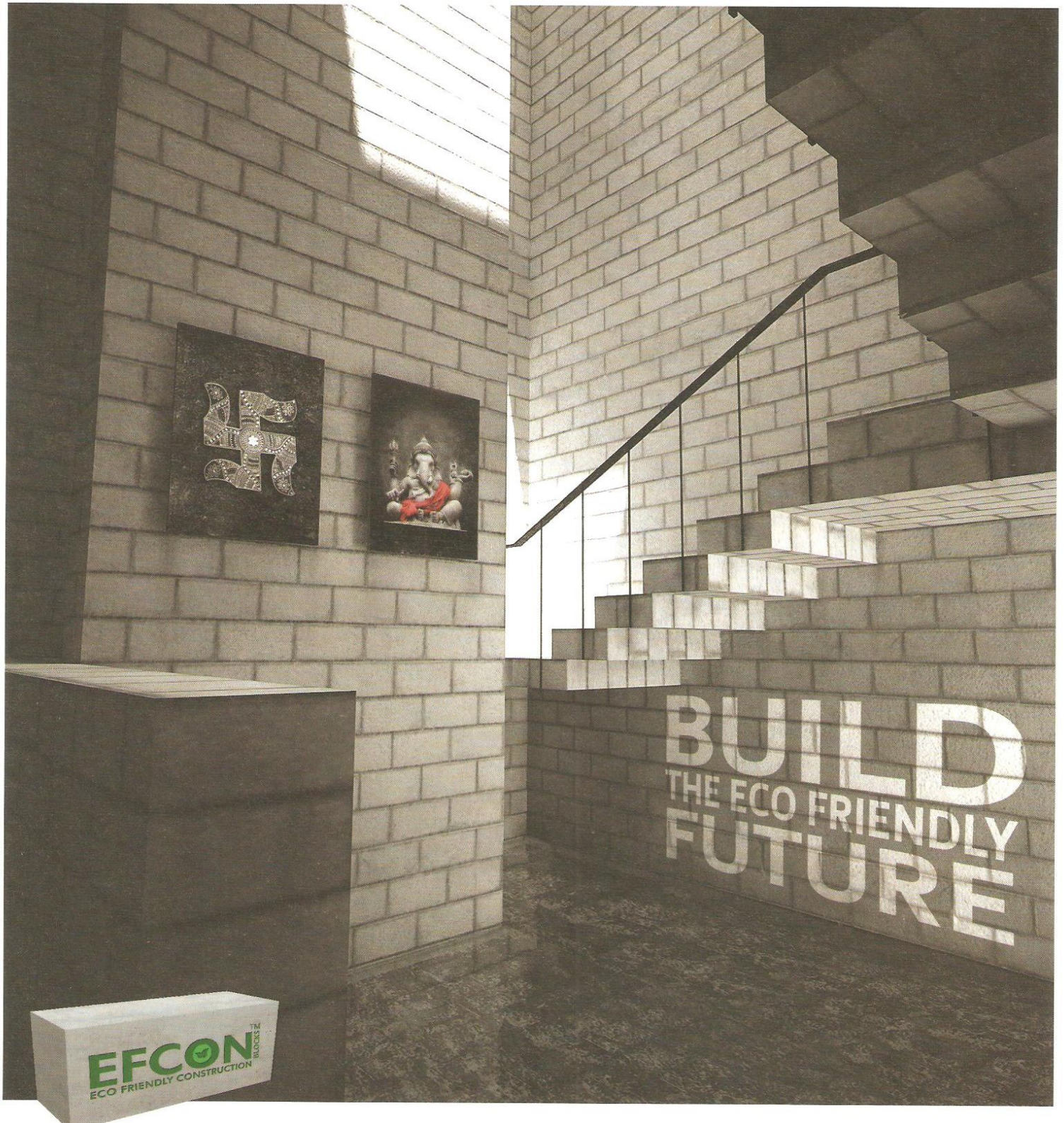
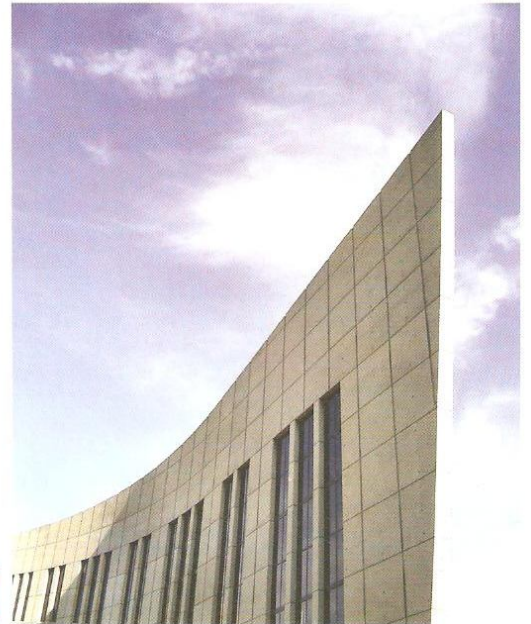






PHENIX



BELIEVES IN GO GREEN CONSTRUCTION

At a time when all industrial sectors are moving towards a 'green' alternative in order to contribute to the environment, here comes EFCON Autoclaved Aerated Concrete (AAC), an Eco-Friendly Construction business "An excellent substitute for the clay bricks & hollow concrete blocks for construction of walls". The construction field is already trending in contributing to the environment, this



Phenix Procon Pvt Ltd is the only company with Advanced technology in entire Gujarat. Phenix furnishes products with good efforts in the market area. It is one of the largest company with Fully Automatic Plant in Gujarat. Phenix Procon Pvt Ltd offers one of the most Eco- friendly, economical and efficient solution in today's construction needs.

Our aim is to provide steady and speedy delivery to our clients. At the earliest orders are dispatched after client's confirmation. In our case variation in supply is not an issue as whole process is mechanized and passes through series of tests.

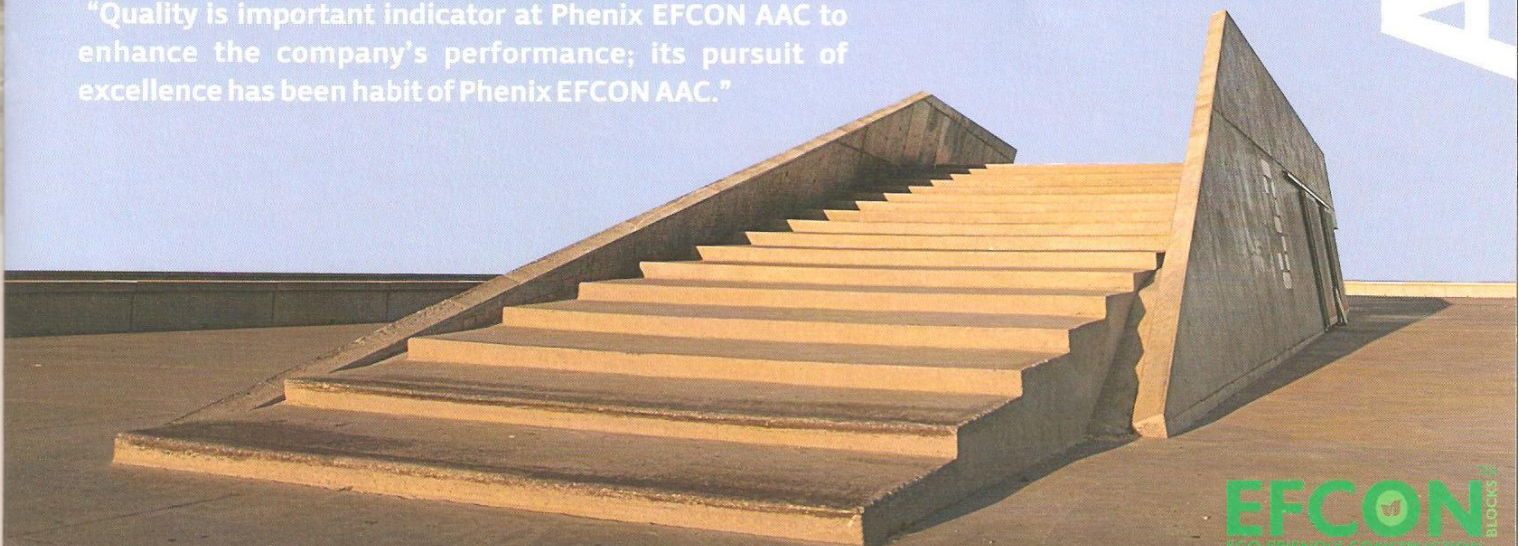
Phenix quality control depends on raw material selection to the production process. In-house test are done at each level of production process to ensure that each and every block is made to perfection. The final products are then delivered to the client.

EFCON blocks are also put through a periodic third party external quality.

"Customer service is not a department, it's everyone's job"

"Quality is important indicator at Phenix EFCON AAC to enhance the company's performance; its pursuit of excellence has been habit of Phenix EFCON AAC."

ABOUT PHENIX



EFCON
BLOCKS
ECO FRIENDLY CONSTRUCTION

AAC ADVANTAGES

Eco-friendly

- Manufacturing process is pollution free.
- Recycled products are used for manufacturing.
- EFCON AAC blocks are low energy consumption.
- EFCON AAC blocks are made up of fly ash wastage materials.
- As EFCON AAC blocks are made up of fly ash, they are Non-Toxic.
- EFCON AAC blocks has very low emission and does not generate any waste products.
- Nowadays, AAC blocks is highly recommended for any 'Green Building' project.



Thermally Insulated & Energy Efficient

- Reduces energy consumption.
- Saving energy costs of air-conditioning.
- Whole wall coverage and low air infiltration.
- Thermal insulation reducing heating in building.
- EFCON AAC blocks offer excellent thermal insulation



Lightweight

- Easy installation due to light weight.
- Easier and cheaper for transportation.
- EFCON AAC blocks lighter than normal bricks.
- Allowing taller buildings construction because of its lightweight.
- Millions of tiny air cells impart EFCON AAC products a very light weight structure.
- Density of these light weight EFCON AAC blocks usually ranges between 400 – 800 kg/m³.



Faster Construction

- Reduces construction time by 20%.
- Lighter and bigger blocks make construction easier and faster.
- Easy to install.
- Sets and hardens quickly.





- Termites and ants do not eat EFCON AAC blocks.
- EFCON AAC blocks cannot be damaged or infested easily.
- As Fly ash or pond ash is an inert material and does not allow termites or other pests to survive.

Earthquake Resistant



- EFCON AAC blocks resist fire, earthquake.
- Low amount mass of a structure due to which the impact of an earthquake on a building is decreasing.

Versatility



- AAC products can be used to make any design, be it carving or irregular shapes.
- It can be manufactured in various shapes and sizes to suit intended purpose.

Sound Insulation



- Tiny air cell structure provides sound absorption qualities.
- Ideal for schools, hospitals, hotels, IT offices, recording studio construction.
- Sound insulation of a wall built using AAC is much higher than a wall of similar thickness built using other materials.

Easy Workability and Design Flexibility



- Custom sizes available.
- Easier electrical installations.
- Easily cut faster than wood using normal tools, drilled, nailed, milled ability.
- AAC products are very easy to work with.
- It does not shrink or warp once it is removed from autoclave.
- AAC is cured using steam, it is ready to be used as soon as it comes out of autoclave.

Cost Effective



- Reduces concrete uses up to 90% then normal clay blocks.
- Saves steel up to 20-25%.
- Phenix EFCON AAC blocks are so cheaper in cost than normal clay blocks.

Fire Resistant



- EFCON AAC blocks are Non-combustible
- EFCON AAC are inorganic and incombustible.
- EFCON AAC is especially suited for fire-rated applications.
- EFCON AAC blocks have higher rate of survivability in case of fire.

AAC

WORKING GUIDELINES

Coping Beam

Do coping beam with 2nos 8mm reinforcement after 1.2mts height.



Don't wall without coping beam.



Max. Height of wall built in a day

Do make sure that there should be 6 layer construction per day.



Don't allow more than 6 layer construction per day.



Block Cutting

Do use tool like hacksaw or rotary cutter.



Don't Cut blocks with hammer or chisel.

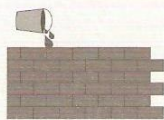


Curing of Masonry Wall

Do curing only for mortar joint.

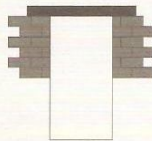


Don't so flooding of wall with water.

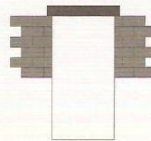


Lintel

Lintel should be rest on full blocks

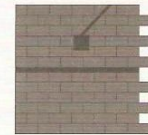


Don't lintel be rest on half blocks



Electrical & Sanitary Chases

Do before plastering depth of chases vertical < 1/3 of wall thickness and horizontal < 1/6 of wall thickness.



Don't do it after plastering.



Chases Area

Do fill with compact mortar & apply wire mesh on it.

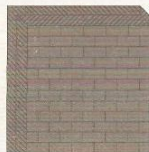


Don't plaster without wire mesh on chases area.



Beam & Column Junction

There should be protection net for column and beam.



Don't start plastering if there is no protection net for column and beam.



Plastering Thickness

Do keep thickness external 15mm and internal 10mm.

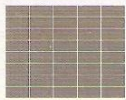


Don't keep thickness more than external 15mm and internal 10mm.



Stacking & Storage

On dry and even surface to avoid damage and contact with moisture



on uneven and wet surface



Mortal For Masonry

Mortar mix 1:6 (Cement : Sand)



1 : 6



Cement Sand

Rich Mortar mix 1:4 (Cement : Sand) will shrink so it leaves bond with block



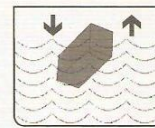
1 : 4



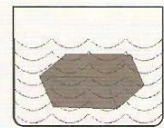
Cement Sand

Wetting of block before Application

deep in water & Lift immediately



Keep it immersed or soak for long time



Mortal Thickness

Keep it limited to 10 to 12mm

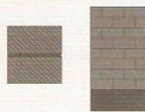


Keep it More than 12mm



Mortal Joints

Recessed joint recommended for effective plaster bonding with masonry



Keep flush surface

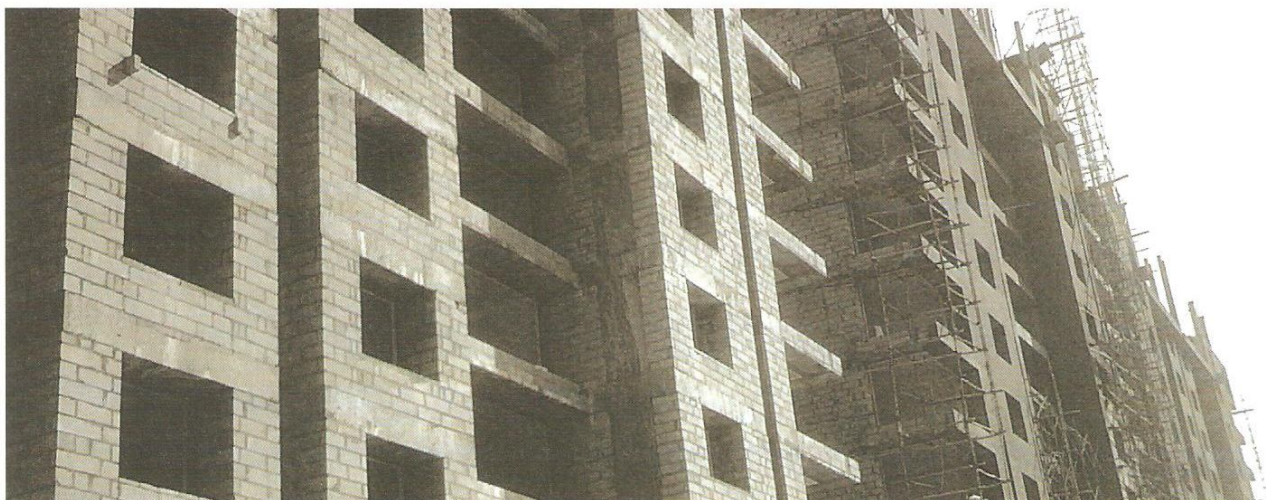


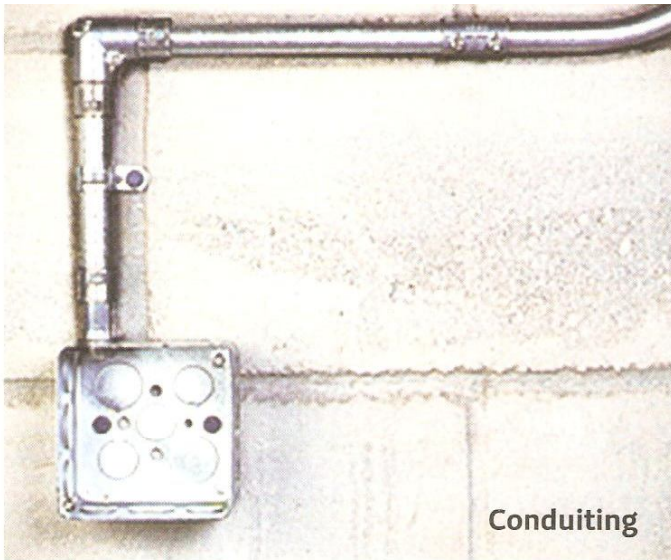
Bond Pattern

Joint to be staggered with minimum 100mm bearing

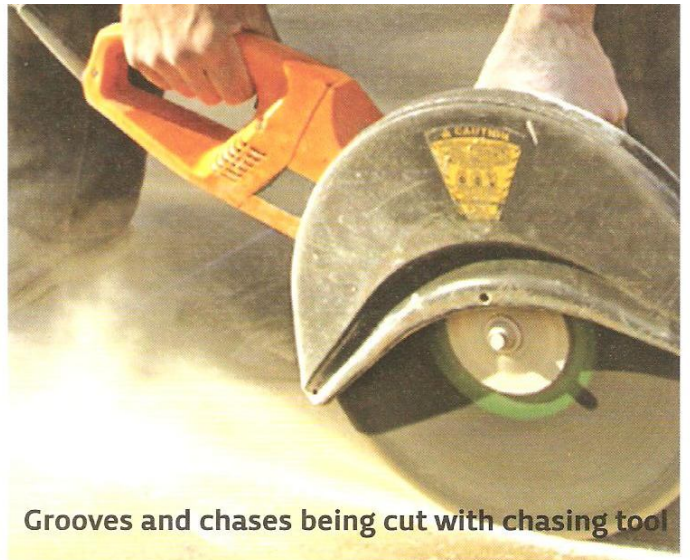


Keep lesser than 100mm bearing

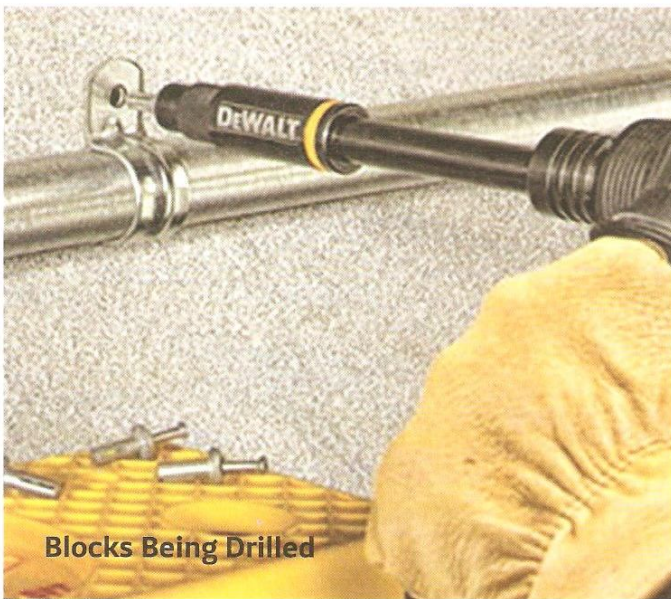




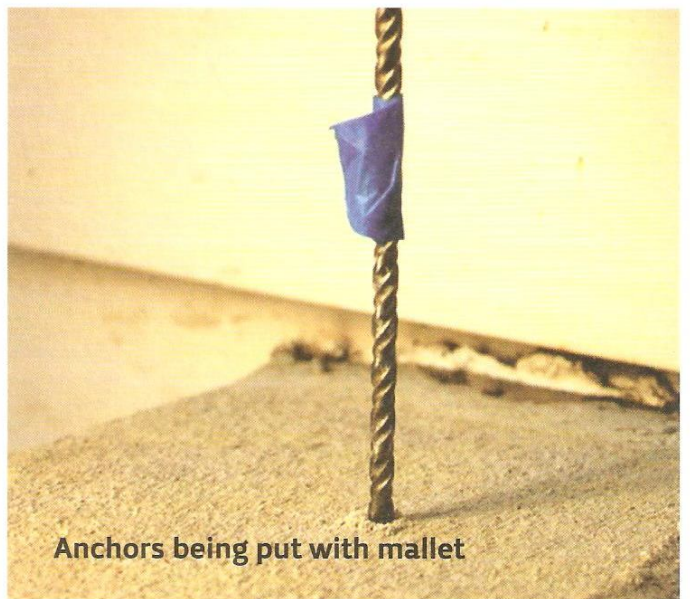
Conduiting



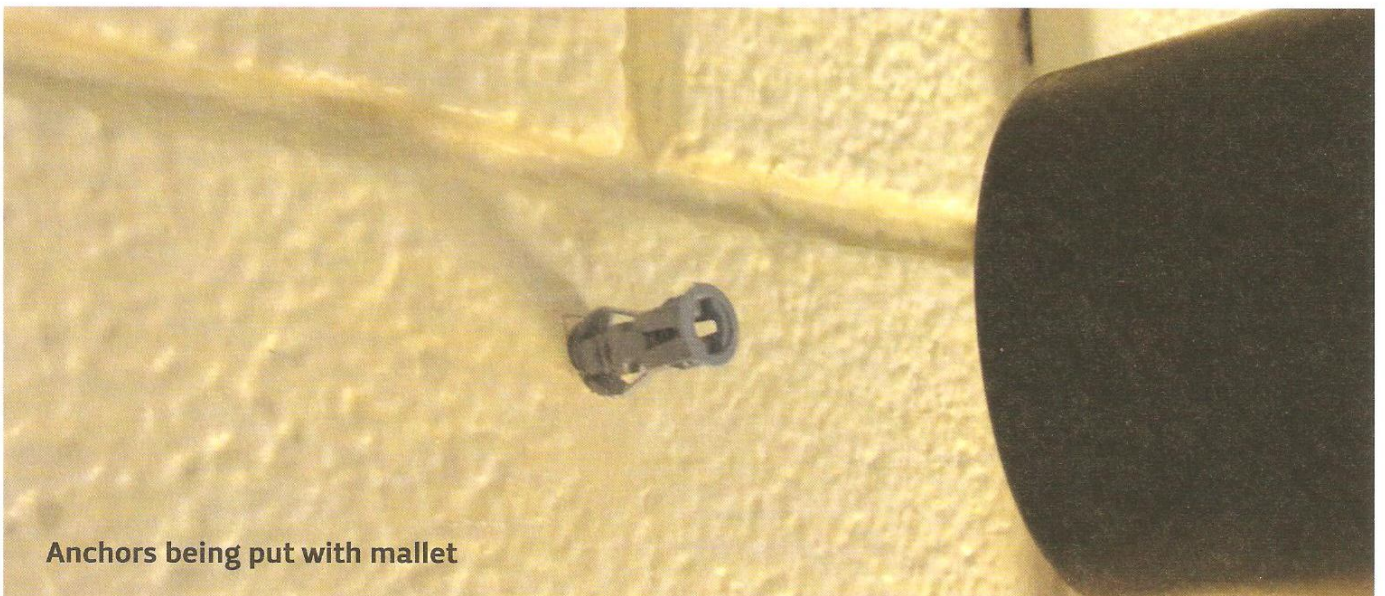
Grooves and chases being cut with chasing tool



Blocks Being Drilled



Anchors being put with mallet



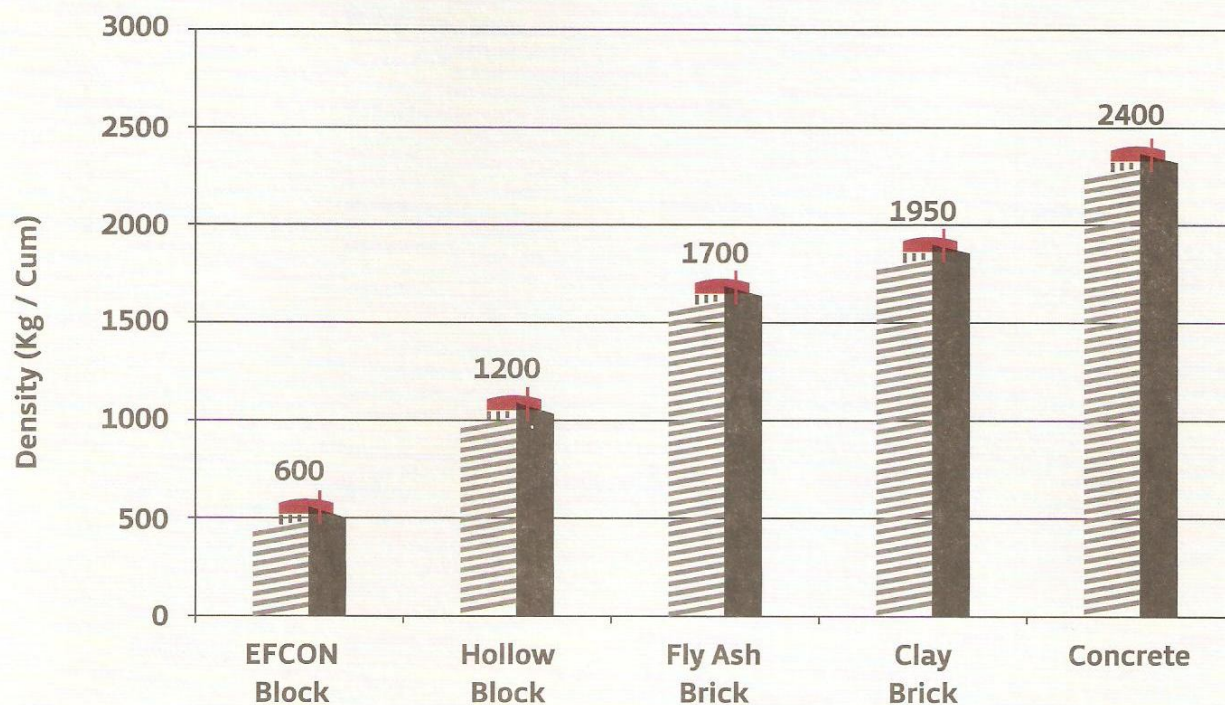
Anchors being put with mallet

AAC

SIZE & SPECIFICATION

No.	Parameters	Unit	Value
01.	Size	MM	600 X 200
		Inch	24 X 8
02.	Thickness	MM	75, 100, 150, 200, 225
		Inch	3, 4, 6, 8, 9
03.	Precision in size	MM	1.5mm (+/-)
04.	Compressive Strength	N / MM ²	As per IS 2185
05.	Dry Density	Kg / Cum	550 to 650 (Oven Dry)
06.	Fire Resistance	Hours	02 to 06 depending upon thickness
07.	Sound Insulation index	db	45 for 200 mm thick wall
08.	Thermal Conductivity	W / (k-m)	0.24
09.	Drying Shrinkage	%	0.04

Density Comparison of Walling Materials



USES OF AAC



Offices



Residencies

Skyscrapers

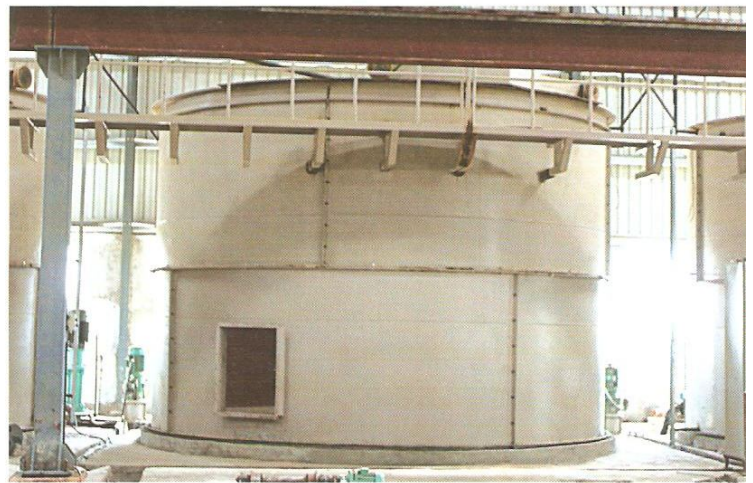
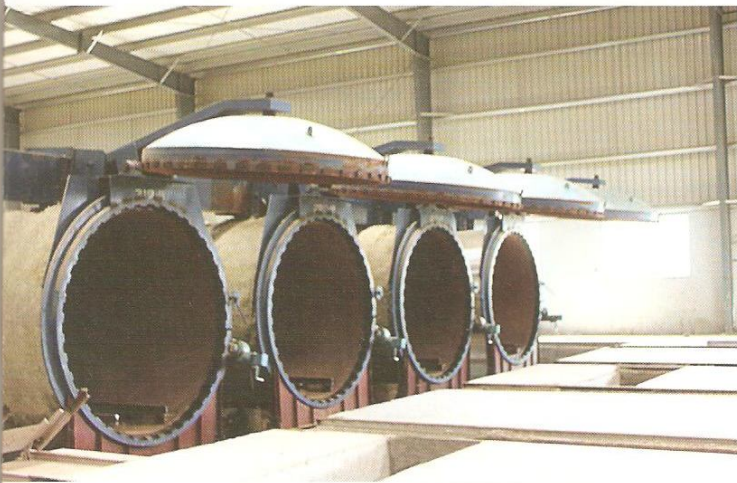


Hospitals

Hotels



Malls



EFCONTM
BLOCKS
ECO FRIENDLY CONSTRUCTION

VISUAL OF
TOUR SITE



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